Failure of proximity gaps close to capacity*

Dmitry Krachun[1], Stepan Kazanin[2], and Ulrich Haböck[3]

[1] Princeton University
[2] Stanford University
[3] StarkWare Industries

Abstract. We give a simple counterexample which shows that, for Reed–Solomon codes over multiplicative subgroups of prime fields, proximity gaps do not hold near capacity, at least not as conjectured by Ben-Sasson, et al., in [BCI+20]. For relative distance $\theta = 1 - \rho - \eta$, where ρ is the rate of the code, and positive $\eta = \Theta_\rho(1/\log n)$, where n is the length of the code, we construct an affine line that is not entirely θ-close to the code but still contains $2^{\Omega_\rho(1/\eta)}$ such points. The same construction gives a slightly stronger list-decoding lower bound. The proof uses a new additive-combinatorics lemma on sums of roots of unity.

1 Introduction

Let $\mathbb{F}$ be a finite field, and $\mathcal{D}$ be a finite subset of $\mathbb{F}$ of size n, and let $C = \mathrm{RS}[\mathbb{F}, \mathcal{D}, k]$ denote the Reed–Solomon code obtained by evaluating polynomials of degree at most k on $\mathcal{D}$. For any two $u, v \in \mathbb{F}^{\mathcal{D}}$, $\Delta(u, v)$ denotes the relative Hamming distance. Given an affine line in $\mathbb{F}^{\mathcal{D}}$, spanned by two words $u_0, u_1 \in \mathbb{F}^{\mathcal{D}}$, where $u_1 \neq 0$,

$$L = \{u_0 + \lambda u_1 \,:\, \lambda \in \mathbb{F}\}\,,$$

and parameter $\theta \in (0, 1)$, the *proximity-gap* question asks whether L can contain many words u with relative distance $\Delta(u, C) = \min_{c \in C} \Delta(u, c) \leq \theta$, without being entirely contained in the θ-neighborhood of C. That is, one asks for an upper bound on the number of θ-close points

$$|\{\lambda \in \mathbb{F} \,:\, \Delta(u_0 + \lambda u_1, C) \leq \theta\}|\,,$$

over all lines which are not entirely θ-close. This upper bound is called *soundness error* of the proximity gap at distance θ.

Proximity gaps play a key role in the construction of interactive oracle proofs [BCS16], also called probabilistic checkable interactive proofs [RRR16], and in particular in the celebrated FRI proof of proximity to Reed–Solomon codes [BSBHR18], which is widely used in practice. A convenient close-to-capacity version of the conjectural picture is the following:

* A preliminary version of this paper has been recently shared with the Ethereum Foundation, in the context of their proximity prize [Fou25].

Conjecture 1 ([BCI+20]). *There exist universal constants $c_1, c_2 > 0$ such that for every finite field $\mathbb{F}$ and every Reed–Solomon code $C = \mathrm{RS}[\mathbb{F}, \mathcal{D}, k]$ of length $n = |\mathcal{D}|$ and distance $\delta = 1 - k/n$, the proximity gaps soundness error at distance $\theta = \delta - \eta$, where $\eta > 0$, is at most*

$$a(\theta) \leq \frac{n^{c_1}}{(\eta(1-\delta))^{c_2}}.$$

Proximity gaps are closely related to list-decodability of the code, and in fact Crites and Stewart [CS25] and Ben-Sasson et al. [BCH+25] recently showed that the former always implies the latter. (We shall dwell on this later on.) List-decodability is of equal importance for the security of interactive oracle proofs, and similar conjectures have been stated over time.

Conjecture 2 ([BGKS20]). *For every $\delta \in (0,1)$, there is a constant $c_\delta > 0$ such that every Reed–Solomon code of length n and relative distance δ is list-decodable from $\theta = \delta - \eta$ fraction errors with list size*

$$\ell(\theta) \leq \left(\frac{n}{\eta}\right)^{c_\delta}.$$

By now there is both positive and negative evidence for the two conjectures. We restrict to the most important results for Reed–Solomon codes, and refer to the introduction of [BCH+25] for a more extensive overview.

- Over general fields there are positive results up to the Johnson radius, namely for $\theta < 1 - \sqrt{1-\delta}$. While list-decodability follows from the classic combinatorial argument, proximity gaps require an involved analysis of the Guruswami–Sudan decoder over algebraic function fields, see [BCI+20] and the recent improvements in [BCH+25].
  An exciting line of work [RW14,ST20,FKS22,GLS+24,BGM23,GZ23,AGL24] shows that random Reed–Solomon codes are optimal list-decodable, i.e. with list sizes $\ell(\theta) = O(\frac{1}{\eta})$, for distances up to the Elias radius[4]

  $$r_E(\rho) = \max\{\theta \,:\, \rho + H_q(\theta) \leq 1\},$$

  which is the information theoretic bound for list-decodability. More recently, Goyal and Guruswami [GG25] show that (among a wider class of codes) Reed–Solomon codes with a random domain also have optimal proximity gaps up to $r_E(\rho)$. That is, $c_1 = 1$ and $c_2 = 1$ in the notation of the above Conjecture 1.
- On the other hand, several recent works show that both conjectures fail in certain settings. Crites and Stewart [CS25] show that Conjecture 1 cannot hold above the Elias radius, since list sizes grow exponentially in n beyond $r_E(\rho)$.

[4] Here, $H_q(\theta) = \theta \log_q(q-1) - \theta \log_q(\theta) - (1-\theta)\log_q(1-\theta)$ is the q-ary entropy function.

Over binary fields, it is known since [BSKR06] that certain Reed–Solomon codes have bad list-decoding properties way before the Elias radius. These examples are explicit, and have been substantially refined in [BCH+25], showing that we also have bad proximity gaps in that range.

The purpose of this note is to demonstrate that for Reed–Solomon codes over smooth multiplicative subgroups of prime fields $\mathbb{F} = \mathbb{F}_p$, both Conjecture 1 and Conjecture 2 fail near capacity. For arbitrarily large $\tau \geq 1$ and $\delta \in (0,1)$, we construct examples of increasing length n in which at distance

$$\theta = \delta - \eta, \quad \eta = \Theta(1/\log n),$$

both the list sizes as well as the proximity gaps soundness error are at least as large as n^τ. (Since the smallest value for η is $1/n$, this in fact invalidates the conjectures.)

Theorem 1. *For $\tau \geq 1$, $\delta \in (0,1)$, and $\beta > \max\{\tau + 1, \frac{12}{5}\}$. There exist infinitely many $n = 2^b$ and primes $p \equiv 1 \bmod n$, $p = \Theta(n^\beta)$, together with Reed–Solomon codes $C = \mathrm{RS}[\mathbb{F}_p, H, k]$ with distance $\delta^* = \delta + o(1/\log n)$ and evaluation domain $H \subset \mathbb{F}_p^\times$, a multiplicative subgroup of size n, so that the following holds for*

$$\eta = \frac{c}{\tau \log n} \cdot (1 + o(1)),$$

where $c \in (0,1)$ is a constant which depends only on δ^*.

1. *There exists a function $u \in \mathbb{F}_p^H$ with*

$$\ell(\delta^* - \eta) = |\{c \in C \,:\, \Delta(c,u) \leq \delta^* - \eta\}| \geq 2^{\frac{c}{\eta}} = n^{\tau - o(1)}.$$

2. *There exist functions $u_0, u_1 \in \mathbb{F}_p^H$ with*

$$a(\delta^* - 2\eta) = |\{\lambda \in \mathbb{F}_p \,:\, \Delta(u_0 + \lambda u_1, C) \leq \delta^* - 2\eta\}| \geq 2^{\frac{c}{\eta}} = n^{\tau - o(1)},$$

and yet $\Delta(u_1, C) \geq \delta^* - \eta$.

In the theorem, the lower bound $\frac{12}{5}$ on β arises from a quantitative version of the Prime Number Theorem in arithmetic progressions, specifically, we use the recent improvement of Thorner and Zaman [TZ24] for the case of smooth moduli.[5] From above, the field size in our construction is unconstrained, and choosing β sufficiently large always produces parameters which fall below the Elias radius.

[5] By the Montgomery conjecture [Mon73] the Prime Number Theorem in arithmetic progressions (i.e. the fact that $\pi(x;n,a) \sim \pi(x)/\varphi(n)$, where $\pi(x;n,a)$ counts the number of primes up to x which are congruent to a modulo n) should hold uniformly in $x \geq n^{1+\varepsilon}$ for any fixed ε. As a consequence, one may expect that in the theorem $\frac{12}{5}$ can be replaced with 1 and so any $\beta > \tau + 1$ works.

Overview.

This note is a polished and extended version of a previous draft, which has been shared with the Ethereum Foundation in regards to their *1 Million Dollar Proximity Prize* [Fou25]. We fill in some details and elaborate on list-decodability (Item 1 above), an aspect that has not been discussed in the draft. The writeup is organized as follows:

- Section 2 covers the original proof of the proximity gap Theorem 1, Item 2, for a slightly larger bound on β than stated above. The construction goes along the same lines as in [BCH+25, Section 7], and is based on vanishing polynomials of large cosets of the domain. However, instead of relying on a conjecture on the sum-sets of multiplicative subgroups, we prove a weaker variant of their conjecture, which is yet sufficient to yield the claimed proximity gap.
- Section 3 then shows how to easily read off the claimed list-decoding center (Item 1 above) from the same construction.
- In Appendix A, we supply a simpler proof of the theorem, based on the above mentioned quotienting construction from [CS25] and [BCH+25]. That construction is generic and highlights the important role of list-decodability. Somewhat surprisingly we are able to yield even slightly better concrete bounds than with the direct approach from Section 2.

We conclude by noting that Theorem 1 affects the parameters of practical interactive oracle proofs more significantly than suggested by Fenzi and Sanso [FS25]. For a detailed discussion of the issue we refer to the whitepaper of Carmon, et al. [CGH+26].

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Related Work

Proximity gaps are closely related to the stronger notion of *correlated agreement* [BSBHR18,BSKS18], stating a joint set on which each word of the line agrees with a codeword. The core strategy behind every positive result in literature, starting with the early results on general linear codes [AHIV17,RZ18] [BSKS18,BGKS20], as well as ones for Reed–Solomon codes in [BCI+20,BCH+25], is to show that a fraction of the close codewords are collinear. This property is dubbed *line decodability* in [GG25], or "collinearity of proximates" in [Hab24], where it is used for the soundness analysis of interactive proofs of proximity, in which the close polynomials are subject to certain constraints. For the same reasons, Zeilberger [Zei24] and Arnon, Chiesa, Fenzi and Yogev [ACFY25] use a stronger, "global" notion of correlated agreement, called *mutual correlated agreement* in [ACFY25]. The correlated agreement results from

[BSKS18,BGKS20] and [BCI+20] have been generalized to mutual correlated agreement in [GKL24,Zei24] and [Hab25], respectively.

Concurrent to [CS25], Diamond and Gruen [DG25] came up with a proximity gap counter example near capacity, yet they did not connect their techniques with the information-theoretic limit.

2 A bad line from vanishing polynomials

In this section, we describe how to explicitly construct a proximity gap example as claimed in Theorem 1, Item 2, with only minor restrictions on the parameter β. The smaller bound on the latter is discussed in Remark 3.

Proposition 1. *For $\tau \geq 1$, $\delta \in [1/2, 1)$, and $\beta > 2\tau + 1$. There exist infinitely many $n = 2^b$ and primes $p \equiv 1 \bmod n$, $p = \Theta(n^\beta)$, together with Reed–Solomon codes $C = \mathrm{RS}[\mathbb{F}_p, H, k]$ with evaluation domain H a multiplicative subgroup of size n, and distance δ^*, so that $|\delta^* - \delta| \leq \frac{3}{2\tau \log n}$ and the following holds for*

$$\eta \sim \frac{c}{\tau \log n}, \quad c = \rho + \frac{1}{2} H_2(2\rho),$$

where $\rho := 1 - \delta$. There exist functions $u_0, u_1 \in \mathbb{F}_p^H$ with

$$|\{\lambda \in \mathbb{F}_p \ : \ \Delta(u_0 + \lambda u_1, C) \leq \delta^* - 2\eta\}| \geq 2^{\frac{c - o(1)}{\eta}} = n^{\tau - o(1)},$$

and yet $\Delta(u_1, C) \geq \delta^* - \eta$.

Remark 1. By taking complements of subsets in our construction below, Proposition 1 holds also for any $\delta \in (0, 1/2)$, with $c = (1 - \rho) + \frac{1}{2} H(2(1 - \rho))$.

2.1 The construction.

The approach is the same as in [BCH+25, Section 7] and based on vanishing polynomials of union of large cosets in H. For constructing sparse polynomials of the desired properties, we use the projection

$$\pi : H \longrightarrow G, \quad x \mapsto x^m,$$

onto a smaller subgroup G of logarithmic size $|G| = s$ which is about $\log |H| = \log n$. Any vanishing polynomial of a subset $S \subset G$,

$$v_S(X) = \prod_{x \in S} (X - x),$$

gives rise to an m-gap polynomial

$$v_S(\pi(X)) = X^{r \cdot m} + c_1 \cdot X^{(r-1) \cdot m} + O(X^{(r-2) \cdot m}),$$

with $r = |S|$. The claimed functions u_0, u_1 correspond to the two leading monomials,

$$u_0(x) = x^{rm}, \quad u_1(x) = x^{(r-1)m},$$

for $x \in H$, and the Reed–Solomon code C is generated by the polynomials

$$p(X) \in \mathbb{F}_p[X]^{\leq (r-2)m}.$$

Note that $u_0 + c_1 u_1$ agrees with a codeword on $|\pi^{-1}(S)| = rm$ points, which is m points more agreement than In terms of relative distances,

$$\Delta(u_0 + c_1 u_1, C) \leq \delta^* - \frac{2}{s} \quad \text{but} \quad \Delta(u_1, C) \geq \delta^* - \frac{1}{s},$$

where $\delta^* = 1 - (r-2)/s$ is the distance of the code C. The main observation is that the leading non-trivial coefficient has an exact expression

$$c_1 = -\sum_{x \in S} x,$$

which allows to relate the number of such linear combinations with the size of certain sum-sets of multiplicative groups.

We claim the following additive combinatorics lemma.

Lemma 1. *Let G be a multiplicative subgroup of a prime field $\mathbb{F}_p$, of size s, a power of two. If $p > s^{s/2}$, then for any integer $1 \leq r \leq s/2$,*

$$|\{x_1 + \ldots + x_r \ : \ \textit{distinct } x_1, \ldots, x_r \in G\}| \geq 2^r \cdot \binom{s/2}{r}.$$

Before proving Lemma 1 we demonstrate how it implies the claimed example, if one does not care about the field size p: For any $\rho := 1 - \delta \in (0, 1/2]$ and $\tau \geq 1$. Choose arbitrary large $s = 2^b$ with integer $r < s/2$ so that

$$\delta^* = 1 - \frac{r-2}{s}$$

satisfies $|\delta^* - \delta| < 1/s$. Take n the unique power of two, so that

$$\frac{1}{2} \cdot 2^{\frac{c}{\tau} \cdot s} < n \leq 2^{\frac{c}{\tau} \cdot s},$$

with $c = c(\delta^*) = \rho + \frac{1}{2} H(2\rho)$ as claimed in Proposition 1. Except for small s, n is larger than s, and in particular $s | n$. With this choice,

$$\frac{c}{\tau} \cdot s - 1 < \log n \leq \frac{c}{\tau} \cdot s,$$

$$\frac{c}{\tau} \cdot \frac{1}{1 + \log n} < \frac{1}{s} \leq \frac{c}{\tau} \cdot \frac{1}{\log n}.$$

Now, by Dirichlet's theorem there are infinitely many primes

$$p \equiv 1 \bmod n,$$

and we can choose any prime p which is larger than $s^{s/2}$. For that p, the prime field $\mathbb{F}_p$ contains a multiplicative subgroup H of size $|H| = n$, which contains a multiplicative subgroup G of size s. We thus can take u_0, u_1 and C over H as above, and use Lemma 1 to conclude that at least

$$a = 2^r \cdot \binom{s/2}{r}$$

linear combinations of u_0 and u_1 are as close to C as desired. By Stirling's formula,

$$a \sim 2^{s(\rho + \frac{1}{2}H_2(2\rho) - o(1))} = 2^{s\cdot(c-o(1))} = 2^{\frac{c-o(1)}{\eta}}.$$

By our choice of s the latter is equal to $n^{\tau+o(1)}$. This shows Proposition 1 for primes $p > s^{s/2}$.

2.2 Proof of Lemma 1

The key idea is that the first $\varphi(s) = s/2$ powers of an s-th primitive root of unity are linear independent over $\mathbb{Q}$, and this independence is preserved "modulo" large primes p, that is when substituting the root with a generator g of G.

We consider *specific* sums of r elements from G: Those which stem from subsets $S \subset G$ of size $|S| = r$ with $S \cap (-S) = \varnothing$. Since $-g^i = g^{s/2+i}$, each such sum corresponds to a *sum-polynomial*, that is a polynomial

$$P(X) = (-1)^{\varepsilon_1} X^{i_1} + \ldots + (-1)^{\varepsilon_r} X^{i_r} \in \mathbb{Z}[X], \tag{1}$$

with

$$(\varepsilon_1, \ldots, \varepsilon_r) \in \{0,1\}^r \text{ and distinct } i_1, \ldots i_r \in [0, s/2). \tag{2}$$

We show that each sum-polynomial yields different values at $X = g$, proving the claim of the lemma.

Let $\Phi_s(X)$ be the s-th cyclotomic polynomial.[6] If any two sum-polynomials $P(X)$, $Q(X)$ have the same value at $X = g$, then

$$\mathsf{Res}(P(X) - Q(X), \Phi_s(X)) \equiv 0 \mod p$$

since both $R(X) = P(X) - Q(X)$ and $\Phi_s(X)$ have a common zero in $\mathbb{F}_p$. Let us switch to the complex numbers to estimate the absolute value of the resultant. By the multiplicativity,

$$\mathsf{Res}(R(X), \Phi_s(X)) = \prod_{\Phi_s(z)=0} \mathsf{Res}(R(X), X - z) = \prod_{\Phi_s(z)=0} R(z),$$

[6] Recall that $\Phi_s(X)$ has integer coefficients, $\deg \Phi_s(X) = \varphi(s) = s/2$, and is irreducible over the rational numbers $\mathbb{Q}$.

where the product goes over all $\varphi(s)$ primitive s-th roots of unity in the complex plane. We thus obtain

$$|\mathsf{Res}(R(X), \Phi_s(X))| = \prod_{\Phi_s(z)=0} |R(z)| \leq (2r)^{\varphi(s)} \leq s^{s/2}, \tag{3}$$

and by the size of p, we must have

$$\mathsf{Res}(R(X), \Phi_s(X)) = 0.$$

Since $\Phi_s(X)$ is irreducible over the rationals, and $\deg P(X) < \varphi(s)$, we therefore must have $R(X) = 0$, in other words, $P(X) = Q(X)$. This completes the proof.

Remark 2. Another way of seeing this proof is to consider the norm on the ring of integers $\mathbb{Z}[\zeta_s]$ of the cyclotomic field $\mathbb{Q}[\zeta_s]$, where ζ_s is an s-th primitive root of unity. If there exists a primitive root $g \in \mathbb{F}_p^*$ and a polynomial $R(X) \in \mathbb{Z}[X]$ such that $R(g) = 0$ in $\mathbb{F}_p$ then the ring homomorphism $\Psi : \mathbb{Z}[\zeta_s] \to \mathbb{F}_p$ mapping ζ_s to g sends $R(\zeta_s)$ to $R(g) = 0$. Thus, Ψ also sends $\mathrm{N}_{\mathbb{Q}[\zeta_s]/\mathbb{Q}}(R(\zeta_s))$ to zero and since $\mathrm{N}_{\mathbb{Q}[\zeta_s]/\mathbb{Q}}(R(\zeta_s)) \in \mathbb{Z}$ we have that p must divide this norm. We have

$$\mathrm{N}_{\mathbb{Q}[\zeta_s]/\mathbb{Q}}(R(\zeta_s)) = \prod_{\Phi_s(z)=0} R(z),$$

and the rest of the proof is identical.

2.3 Polynomial field sizes.

To achieve $p = O(n^\beta)$ as claimed in Proposition 1, we tweak the proof of Lemma 1, by excluding the prime factors of $\mathsf{Res}(P(X) - Q(X), \Phi_s(X))$ of each pairwise difference of sum-polynomials. The number of primes in arithmetic progressions are expected to have density

$$\frac{|\{p \in [x, 2x] \,:\, p \text{ is prime}, p \equiv 1 \bmod n\}|}{x} \sim \frac{1}{\varphi(n)\log(x)}, \tag{4}$$

for large x, ensuring not much less than $n^{\beta-1}$ such primes in the interval $[n^\beta, 2n^\beta]$. By taking β sufficiently large, we ensure that these are more than those primes which stem from the resultants.

To make the heuristic concrete, we use the recent refinement of Thorner and Zaman on the prime number densities in arithmetic progressions. The following theorem is covered by Corollary 3.1 in [TZ24], and the subsequent discussion therein.

Theorem ([TZ24]). *For any smooth integer n (in particular a power of two), and arbitrary small $\varepsilon > 0$. Then over intervals $[x - h, x]$, with $\frac{h}{\varphi(n)} \geq x^{\frac{7}{12}+\varepsilon}$,*

$$\sum_{\substack{x-h<p\leq x,\\ p\equiv d \bmod n}} \log p = \frac{h}{\varphi(n)}(1 + o(1)),$$

as $x \to \infty$, for each of the $\varphi(n)$ coprime residue classes modulo n.

We note that Equation 4 follows from the theorem by partial summation[7], for $x > n^{\frac{12}{5}+\epsilon}$ with some arbitrarily small $\varepsilon > 0$. We are now able to prove:

Lemma 2. *For $\beta > \max\{2\tau + 1, \frac{12}{5}\}$. If s is taken sufficiently large in our construction from Section 2.1, then there exists a prime $p \in [n^\beta, 2n^\beta]$ so that*

$$\mathsf{Res}(P(X) - Q(X), \Phi_s(X)) \not\equiv 0 \mod p,$$

for every pair of distinct sum-polynomials $P(X)$, $Q(X)$ as defined in the proof of Lemma 1.

Proof. By inequality (3) the resultant of each such $P(X) - Q(X)$ has at most

$$\frac{s \log s}{2\beta \log n} = O(\log \log n)$$

prime factors which are larger than n^β. There are $a \sim n^{\tau+o(1)}$ such sum-polynomials, yielding a total of at most

$$a^2 \cdot \frac{s \log s}{2\beta \log n} = n^{2\tau+o(1)}$$

prime factors which are larger than n^β. By the above quantitative version of the Prime Number Theorem in arithmetic progressions, we have

$$\left|\left\{p \in [n^\beta, 2n^\beta] \,:\, p \text{ is prime}, p \equiv 1 \bmod n\right\}\right| \sim \frac{n^\beta}{\varphi(n) \cdot \log(n^\beta)} = \Omega(n^{\beta-1-o(1)}),$$

which by our choice of β is larger than $n^{2\tau+o(1)}$, when n tends to infinity. If n is sufficiently large, we therefore find a prime $p \in [n^\beta, 2n^\beta]$ with the claimed property. □

With the prime p from Lemma 2, our above construction yields as before the bad proximity gap as claimed in Proposition 1.

Remark 3. Lemma 2 can be improved to $\beta > \max\{\tau + 1, \frac{12}{5}\}$ by allowing that few resultants may be zero modulo p. For any small $\varepsilon > 0$: Since

$$\tfrac{n^\beta}{\varphi(n) \cdot \log n} \gg a \cdot \tfrac{s \log s}{2\beta \log n},$$

we can choose a prime $p \in [n^\beta, n^{2\beta}]$, $p \equiv 1 \bmod n$, for which at most $\varepsilon \cdot a$ resultants are divisible by p. (Otherwise we would have more than $a^2 \frac{s \log s}{2\beta \log n}$ primes stemming from such resultants, which is impossible.) For this prime p, remove those sum-polynomials which are involved in such a divisible resultant. (There are at most twice as many, $\leq 2\varepsilon a$.) This shows that there is $\mathcal{S}$ a subset of sum-polynomials, of size $|\mathcal{S}| \geq (1 - 2\varepsilon) \cdot a$, so that $\mathsf{Res}(P(X) - Q(X), \Phi_s(X)) \not\equiv 0 \bmod p$ for every distinct choice $P(X), Q(X) \in \mathcal{S}$.

[7] This standard step can be looked up in any standard textbook on analytic number theory, such as [IK04].

3 Tweaking the construction for list sizes

The construction from Section 2 can be adapted easily, so that one sees that $u_1 = x^{(r-1)m}$ has as many close codewords as claimed in Theorem 1, Item 1.

Note that a vanishing polynomial from a subset $S' \subset G$ of size $r-1$ (instead of r), gives rise to an m-gap polynomial

$$v_{S'}(\pi(X)) = X^{(r-1)m} + \underbrace{c_1' \cdot X^{(r-2)m} + O(X^{(r-3)m})}_{p_{S'}(X)},$$

with $c_1' = -\sum_{S'} x$, and each different sum c_1' yields a different polynomial $p_{S'}$ from the code C, so that

$$\Delta(u_1, p_{S'}) = \delta^* - \frac{1}{s}.$$

We thus obtain the following result in the same way, using Lemma 2.

Proposition 2. *For $\tau \geq 1$, $\delta \in [1/2, 1)$, and $\beta > \max\{2\tau + 1, \frac{12}{5}\}$. Then in the construction from Section 2,*

$$\ell(\delta^* - \eta) = |\{c \in C \,:\, \Delta(c, u_1) \leq \delta^* - \eta\}| \sim 2^{\frac{c - o(1)}{\eta}} = n^{\tau - o(1)}.$$

Remark 4. Using Remark 3, also Proposition 2 can be improved to $\beta > \max\{\tau + 1, \frac{12}{5}\}$.

Remark 5. After finishing the writeup, we stumbled across the fact that Proposition 2 can be proven much simpler: We do not need any additive combinatorics to see that different subsets S' yield different polynomials $p_{S'}(X)$. This follows from that different subsets have different vanishing polynomials! We therefore obtain

$$\binom{s}{r-1} \sim 2^{s(H_2(\rho) - o(1))}$$

many proximate codewords, without additional argument. Since no resultants are involved, the discussion from Section 2.3 can be even relaxed to $\beta > \max\{\tau + 1, \frac{12}{5}\}$. We start from this observation in our simpler proof of Theorem 1 in Appendix A.

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A A simpler proof of Theorem 1

In this section we show how to prove Theorem 1 without Lemma 1. The underlying technique is taking quotients at a random point, which has its root in DEEP algebraic linking [BGKS20], a widely applied method for turning algebraic interactive oracle proofs into proofs of proximity. That it is also useful for examples of bad proximity gaps was only recently discovered by Crites and Stewart [CS25] and Ben–Sasson, et al. [BCH+25]. Both show how to construct a line with a bad proximity gap from a bad list decoding center, thereby connecting proximity gaps with list-decodability.

The principle is as follows. Given a word $u \in \mathbb{F}_p^D$ with a large list $\mathcal{L}$ of close polynomials from the (Reed–Solomon) code C. Then for any point $z \in \mathbb{F}_p$, not necessarily in D,

$$\frac{u(x) - v}{x - z}$$

is again close to a polynomial, if and only if v is the value of one of the proximates of u at the point z. (We ignore the slight change in degree in this simplified explanation.) Thus, if we have a point z which distinguishes the many polynomials in $\mathcal{L}$, then we may use the above quotient form to see that

$$u_0(x) = \frac{u(x)}{x - z}, \quad u_1(x) = \frac{1}{x - z}, \quad x \in D,$$

span a line with many proximate linear combination, yet u_1 is far from the code. (It is a deep hole, meaning that it has the maximum possible distance to the code.)

To make the principle amenable we rely on the following lemma, which shows that such a point always exists.

Lemma 3 ([BCI+20]). *Let $\mathcal{L}$ be a set of functions from S to $\mathbb{F}_q$, so that any two elements of $\mathcal{L}$ agree on at most A points. For any $\alpha \in S$ let $\mathcal{L}(\alpha) = \{f(\alpha) : f \in \mathcal{L}\}$. Then*

$$E_{\alpha \in S}\left[|\mathcal{L}(\alpha)|\right] \geq \frac{1}{2} \min\left(|\mathcal{L}|, \frac{|S|}{A}\right).$$

The proof of the lemma uses the second moments of the probability distributions

$$W_\alpha(x) = \frac{1}{|\mathcal{L}(\alpha)|} \left|\{f(\alpha) = x \,:\, f \in \mathcal{L}\}\right|,$$

for estimating the expectation of $|\mathcal{L}(\alpha)| \geq \frac{1}{\sum W_\alpha(x)^2}$. We refer to [BCH+25, Section 6] for details.

The construction

The underlying construction is as in Section 2 and 3, but we directly start off with a bad list-decoding center. That is, H will be a multiplicative subgroup of size $|H| = n$ (a power of two), and G will be smaller subgroup $G \subset H$ of logarithmic size $|G| = s = \Theta(\log n)$. Recall that

$$\pi : H \longrightarrow G, \quad \pi(x) = x^m,$$

denotes the projection which maps H onto G in an m-to-1 manner.

1. For $\rho := 1 - \delta \in (0, 1)$, $\tau \geq 1$, and $\beta > \max\{\tau + 1, \frac{12}{5}\}$. As in Section 2, we choose $s = 2^b$ with integer $r < s$ so that

$$\delta^+ = 1 - \frac{r-1}{s}$$

satisfies $|\delta - \delta^+| < 1/s$, and take $n > s$, again a power of two, so that

$$\frac{\tau \log n}{c} \leq s < \frac{\tau(1 + \log n)}{c}$$

but with $c = c(\delta^*) = H_2(\rho)$.

2. Choose any prime $p \equiv 1 \bmod n$ in the interval $[n^\beta, 2n^\beta]$. (By our choice of β such a prime exists.)

With that prime p, the field $\mathbb{F}_p$ has a multiplicative subgroup H of size n, with a subgroup $G < H$ of size s. Take

$$u(x) = x^{rm}, \quad x \in H,$$

and the code

$$C^+ = \mathrm{RS}[\mathbb{F}_p, H, k^+]$$

being generated by the polynomials of degree at most $k^+ = (r-1)m$. (Again, C^+ has distance δ^+.) As in Section 3, each vanishing polynomial $v_S(X^m)$ from subsets $S \subset G$ of size $|S| = r$ is a member of

$$\mathcal{L} = \left\{ c \in C^+ \ : \ \Delta(c, u) \leq \delta^+ - \frac{1}{s} \right\},$$

and we obtain that

$$|\mathcal{L}| \geq \binom{s}{r} \sim 2^{s(H_2(\rho) - o(1))} = n^{\tau - o(1)}.$$

(The proximate polynomials stem from the vanishing polynomials of the r-subsets of G.) Since $p > n^\beta > n^{\tau+1}$, Lemma 3 guarantees a point $z \in \mathbb{F}_p$ with

$$|\mathcal{L}(z)| \geq \frac{1}{2} n^{\tau - o(1)}.$$

To obtain the proximity gaps example, we do the quotienting construction in a slightly stronger form. The polynomials in $\mathcal{L}$ which stem from the above vanishing polynomials are all of the form $p_S(X^m)$, $\deg p_S(X) \le r-1$, hence they take constant values over cosets of the subgroup of m-th roots of unity. For these polynomials,

$$q_S(X) = \frac{p_S(X^m) - p_S(z^m)}{X^m - z^m}$$

is again a polynomial, and $\deg q_S(X) \le (r-2)m$. We thus take

$$u_0(x) = \frac{u(x)}{x^m - z^m}, \quad u_1(x) = \frac{1}{x^m - z^m}, \quad x \in H, \tag{5}$$

and the code

$$C = \mathrm{RS}[\mathbb{F}_p, H, k] \tag{6}$$

is generated by the polynomials of degree at most $k = (r-2)m$. The following holds:

- We have
$$\Delta(u_1, C) \ge 1 - \frac{r-1}{s} = \delta^* - \frac{1}{s}.$$
This is so, because u_1 can agree with a degree $\le (r-2)m$ poly on at most $(r-1)m$ points.
- For every $\lambda \in \mathcal{L}(z)$,
$$\Delta(u_0 + \lambda u_1, C) \le 1 - \frac{r}{s} = \delta^* - \frac{2}{s}.$$
In fact, take any $p_S(X) \in \mathcal{L}$ which stems from a vanishing polynomial $v_S(X^m)$. The polynomial $p_S(X)$ agrees with u on a set of density $\frac{r}{s}$, and over the same set $u_0 + \lambda u_1$, $\lambda = p_S(z^m)$, agrees with the quotient $q_S(X) \in C$.

We thus arrive at the following proposition.

Proposition 3. *For $\tau \ge 1$, $\delta \in (0,1)$, and $\beta > \max\{\tau + 1, \frac{12}{5}\}$. There exist infinitely many $n = 2^b$ and primes $p \equiv 1 \bmod n$, $p = \Theta(n^\beta)$, together with Reed–Solomon codes $C = \mathrm{RS}[\mathbb{F}_p, H, k]$ with evaluation domain H a multiplicative subgroup of size n, and distance δ^*, so that $|\delta^* - \delta| \le \frac{3}{2\tau \log n}$ and the following holds for*

$$\eta \sim \frac{c}{\tau \log n}, \quad c = H_2(\rho),$$

where $\rho := 1 - \delta$. There exist functions $u_0, u_1 \in \mathbb{F}_p^H$ with

$$|\{\lambda \in \mathbb{F}_p \,:\, \Delta(u_0 + \lambda u_1, C) \le \delta^* - 2\eta\}| \gtrsim 2^{\frac{c - o(1)}{\eta}} = n^{\tau - o(1)},$$

and yet $\Delta(u_1, C) \ge \delta^* - \eta$.

The list-size example for the same code C is as argued as in Section 3. This completes the proof of Theorem 1.

We conclude by citing an extreme variant of Proposition 3 in the spirit of [BCH+25].

Proposition 4. *For any $\delta \in (0,1)$ and $\beta > \frac{12}{5}$. There exist infinitely many $n = 2^b$ and primes $p \equiv 1 \bmod n$, $p = \Theta(n^\beta)$, together with Reed–Solomon codes $C = \mathrm{RS}[\mathbb{F}_p, H, k]$ with evaluation domain H a multiplicative subgroup of size n, and distance δ^*, so that $\delta^* - \delta = O(\frac{1}{\log n})$ and the following holds for $\eta = \Theta(\frac{1}{\log n})$. There exist functions $u_0, u_1 \in \mathbb{F}_p^H$ with*

$$|\{\lambda \in \mathbb{F}_p \,:\, \Delta(u_0 + \lambda u_1, C) \le \delta^* - 2\eta\}| \ge \frac{p}{2n},$$

and yet $\Delta(u_1, C) \ge \delta^* - \eta$.

Proof. In the construction choose any $\tau > \beta - 1$ so that the list size is $\ge \frac{p}{n}$. □